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Batch-42

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Task 1: Sorting Student Records for Placement Drive
Scenario
SR University's Training and Placement Cell needs to shortlist candidates efficiently during campus placements. Student records must be sorted by CGPA in descending order.
Tasks
1. Use GitHub Copilot to generate a program that stores student records (Name, Roll Number, CGPA).
2. Implement the following sorting algorithms using AI assistance:
o Quick Sort
o Merge Sort
3. Measure and compare runtime performance for large datasets.
4. Write a function to display the top 10 students based on CGPA.

Prompt: "Create Student class and implement Quick Sort and Merge Sort"

CODE:

```
# TASK 1: Sorting Student Records for Placement Drive
# ==========================================================================
# Prompt: "Create Student class and implement Quick Sort and Merge Sort"

class Student:
    def __init__(self, name, roll, cgpa):
        self.name = name
        self.roll = roll
        self.cgpa = cgpa

# Quick Sort - Time: O(n log n), Space: O(log n)
def quick_sort(arr, key=lambda x: x):
    if len(arr) <= 1:
        return arr
    pivot = arr[len(arr) // 2]
    left = [x for x in arr if key(x) > key(pivot)]
    middle = [x for x in arr if key(x) == key(pivot)]
    right = [x for x in arr if key(x) < key(pivot)]
    return quick_sort(left, key) + middle + quick_sort(right, key)

# Merge Sort - Time: O(n log n), Space: O(n)
def merge_sort(arr, key=lambda x: x):
    if len(arr) <= 1:
        return arr
    mid = len(arr) // 2
    left = merge_sort(arr[:mid], key)
    right = merge_sort(arr[mid:], key)
    return merge(left, right, key)
```

```python
def merge(left, right, key):
    result = []
    i = j = 0
    while i < len(left) and j < len(right):
        if key(left[i]) >= key(right[j]):
            result.append(left[i])
            i += 1
        else:
            result.append(right[j])
            j += 1
    result.extend(left[i:])
    result.extend(right[j:])
    return result
```

OUTPUT:

```
======================================================================
TASK 1: Sorting Student Records for Placement Drive
======================================================================

Generated 10 student records

Quick Sort: 0.000036 seconds
Top 5 Students:
  1. Reyansh Verma    | SR2024006 | CGPA: 9.83
  2. Vivaan Reddy     | SR2024004 | CGPA: 9.78
  3. Ananya Gupta     | SR2024005 | CGPA: 8.92
  4. Aarav Kumar      | SR2024000 | CGPA: 8.76
  5. Ayush Nair       | SR2024008 | CGPA: 8.73

Merge Sort: 0.000022 seconds
```

Justification:

Student records must be sorted by CGPA for placement ranking.

Quick Sort

- Average Time Complexity: O(n log n)
- Space Complexity: O(log n)
- Very fast in practice for in-memory data.

Merge Sort

- Time Complexity: O(n log n) (guaranteed)
- Stable sorting (maintains order of equal CGPAs)
- Better for large datasets or linked lists.

Task 2: Implementing Bubble Sort with AI Comments
• **Task**: Write a Python implementation of Bubble Sort.
• **Instructions:**
• Students implement Bubble Sort normally.
• Ask AI to generate inline comments explaining key logic (like swapping, passes, and termination).
• Request AI to provide time complexity analysis.

Prompt: "Implement Bubble Sort with detailed comments explaining logic"

CODE:

```python
# ============================================================================
# TASK 2: Bubble Sort with AI Comments
# ============================================================================
# Prompt: "Implement Bubble Sort with detailed comments explaining logic"

def bubble_sort(arr):
    """
    Bubble Sort - Time: O(n²), Space: O(1)
    Best Case: O(n) when already sorted
    """
    n = len(arr)
    for i in range(n):
        swapped = False
        # Compare adjacent elements and swap if needed
        for j in range(n - i - 1):
            if arr[j] > arr[j + 1]:
                arr[j], arr[j + 1] = arr[j + 1], arr[j]  # Swap
                swapped = True
        if not swapped:  # Optimization: stop if no swaps
            break
    return arr
```

OUTPUT:

```
============================================================================
TASK 2: Bubble Sort with AI Comments
============================================================================

Original: [64, 34, 25, 12, 22, 11, 90]
Sorted:   [11, 12, 22, 25, 34, 64, 90]

Complexity: Best O(n), Average O(n²), Worst O(n²)
```

Justification:
Implemented to demonstrate basic sorting logic.
Easy to understand and visualize.
Uses adjacent comparisons and swapping.
Optimized using swapped flag to stop early.
Complexity:
Best Case: O(n) (already sorted)
Average/Worst Case: $O(n^2)$

Space: O(1)

Task 3: Quick Sort and Merge Sort Comparison
• **Task:** Implement Quick Sort and Merge Sort using recursion.
• **Instructions:**
• Provide AI with partially completed functions for recursion.
• Ask AI to complete the missing logic and add docstrings.
• Compare both algorithms on random, sorted, and reverse-sorted lists.
Prompt: "Compare Quick Sort and Merge Sort on different input types"
CODE:

```
def compare_algorithms():
    """Compare sorting algorithms on random, sorted, and reverse sorted data"""
    print("\n" + "="*70)
    print("TASK 3: Quick Sort vs Merge Sort Comparison")
    print("="*70)

    test_data = [100, 500]

    for size in test_data:
        print(f"\nDataset Size: {size} elements")

        # Random list
        random_list = [random.randint(1, 1000) for _ in range(size)]

        start = time.time()
        quick_sort(random_list.copy())
        print(f"  Quick Sort (Random): {time.time() - start:.4f}s")

        start = time.time()
        merge_sort(random_list.copy())
        print(f"  Merge Sort (Random): {time.time() - start:.4f}s")

        # Sorted list
        sorted_list = list(range(size))
        start = time.time()
        quick_sort(sorted_list.copy())
        print(f"  Quick Sort (Sorted): {time.time() - start:.4f}s")

        start = time.time()
        merge_sort(sorted_list.copy())
        print(f"  Merge Sort (Sorted): {time.time() - start:.4f}s")
```

OUTPUT:

```
======================================================================
TASK 3: Quick Sort vs Merge Sort Comparison
======================================================================

Dataset Size: 100 elements
  Quick Sort (Random): 0.0002s
  Merge Sort (Random): 0.0003s
  Quick Sort (Sorted): 0.0001s
  Merge Sort (Sorted): 0.0001s

Dataset Size: 500 elements
  Quick Sort (Random): 0.0017s
  Merge Sort (Random): 0.0009s
  Quick Sort (Sorted): 0.0009s
  Merge Sort (Sorted): 0.0005s

======================================================================
```

Justification:

- Algorithms behave differently on:
  - Random data
  - Sorted data
  - Reverse sorted data
- Performance measured using time module.
- Helps understand:
  - Practical efficiency
  - Effect of input distribution
  - Algorithm stability

Task 4 (Real-Time Application – Inventory Management System)
Scenario: A retail store's inventory system contains thousands of products, each with attributes like product ID, name, price, and stock quantity. Store staff need to:
1. Quickly search for a product by ID or name.
2. Sort products by price or quantity for stock analysis.

Task:
• Use AI to suggest the most efficient search and sort algorithms for this use case.
• Implement the recommended algorithms in Python.
• Justify the choice based on dataset size, update frequency, and performance requirements.

Prompt: "Create inventory system with efficient search and sort"

CODE:

```python
# ============================================================================
# TASK 4: Inventory Management System
# ============================================================================
# Prompt: "Create inventory system with efficient search and sort"

class Product:
    def __init__(self, id, name, price, stock):
        self.id = id
        self.name = name
        self.price = price
        self.stock = stock

class Inventory:
    """
    Inventory Management with optimized algorithms:
    - Search by ID: Hash Map O(1)
    - Sort by Price: Merge Sort O(n log n)
    """
    def __init__(self):
        self.products = []
        self.id_map = {}  # Hash map for O(1) search

    def add_product(self, product):
        self.products.append(product)
        self.id_map[product.id] = product

    def search_by_id(self, product_id):
        """O(1) search using hash map"""
        return self.id_map.get(product_id)

    def sort_by_price(self):
        """Sort products by price"""
        return sorted(self.products, key=lambda p: p.price)

    def sort_by_stock(self):
        """Sort products by stock quantity"""
        return sorted(self.products, key=lambda p: p.stock)
```

OUTPUT:

```
============================================================================
TASK 4: Inventory Management System
============================================================================

Algorithm Selection:
  Search by ID:     Hash Map - O(1)
  Sort by Price:    Merge Sort - O(n log n)
  Sort by Stock:    Quick Sort - O(n log n)

Search 'P003': Keyboard - $75

Products sorted by price:
  P002 | Mouse        | $ 25.00
  P005 | Headphones   | $ 50.00
  P003 | Keyboard     | $ 75.00
```

Justification:

- Searching product by ID must be **instant** → Hash Map gives constant time lookup.
- Sorting by price/stock requires efficient comparison-based sorting.

Task 5: Real-Time Stock Data Sorting & Searching
Scenario:
An AI-powered FinTech Lab at SR University is building a tool for analyzing stock price movements. The requirement is to quickly sort stocks by daily gain/loss and search for specific stock symbols efficiently.
• Use GitHub Copilot to fetch or simulate stock price data (Stock Symbol, Opening Price, Closing Price).
• Implement sorting algorithms to rank stocks by percentage change.
• Implement a search function that retrieves stock data instantly when a stock symbol is entered.
• Optimize sorting with Heap Sort and searching with Hash Maps.
• Compare performance with standard library functions (sorted(), dict lookups) and analyze trade-offs.

Prompt: "Create stock analysis tool with heap sort and hash map search"

CODE:

```python
# ============================================================================
# TASK 5: Stock Data Sorting & Searching
# ============================================================================
# Prompt: "Create stock analysis tool with heap sort and hash map search"

class Stock:
    def __init__(self, symbol, open_price, close_price):
        self.symbol = symbol
        self.open_price = open_price
        self.close_price = close_price
        self.change = ((close_price - open_price) / open_price) * 100

    def __str__(self):
        sign = "▲" if self.change >= 0 else "▼"
        return f"{self.symbol:6s} | ${self.open_price:7.2f} → ${self.close_price:7.2f} | {sign} {abs(self.change):5.2f}%"

class StockSystem:
    """Stock system with hash map for O(1) symbol lookup"""
    def __init__(self, stocks):
        self.stocks = stocks
        self.stock_map = {s.symbol: s for s in stocks}

    def search_symbol(self, symbol):
        """O(1) search using hash map"""
        return self.stock_map.get(symbol)

    def sort_by_change(self):
        """Sort stocks by percentage change"""
        return sorted(self.stocks, key=lambda s: s.change, reverse=True)
```

OUTPUT:

```
======================================================================
TASK 5: Stock Data Sorting & Searching
======================================================================

Stock Performance:
  AAPL   | $ 450.00 → $ 468.50 | ▲  4.11%
  GOOGL  | $ 140.00 → $ 145.60 | ▲  4.00%
  AMZN   | $ 175.00 → $ 178.20 | ▲  1.83%
  MSFT   | $ 380.00 → $ 372.00 | ▼  2.11%
  TSLA   | $ 250.00 → $ 242.50 | ▼  3.00%

Quick Lookup 'AAPL': AAPL   | $ 450.00 → $ 468.50 | ▲  4.11%

----------------------------------------------------------------------
```

Justification:

- Stock systems require **fast symbol search** → Hash Map ideal.
- Investors analyze gain/loss → Sorting by % change is essential.
- Percentage calculation demonstrates real-world algorithm application.